25 Years of SMOTE: What Has Been Done and How Did It Benefit Society?

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Background: The Synthetic Minority Over-sampling Technique (SMOTE), introduced in this journal in 2002, has become one of the most widely used remedies for class imbalance in machine learning. [TODO: One or two sentences on why a 25-year retrospective is needed now.]
Objectives: This survey (i) maps and categorizes the SMOTE family of methods (variants, hybrids, and successors) developed since 2002, (ii) documents where SMOTE-based methods have been applied and the benefits reported, and (iii) identifies open problems that motivate future oversampling methods.
Methods: [TODO: Search strategy, databases, time window, inclusion and exclusion criteria, PRISMA-style screening.]
Results: [TODO: Number of studies reviewed, the resulting taxonomy, and the main findings on binary and multiclass settings and on applications.]
Conclusions: [TODO: Main take-away messages and future directions.]

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1 Introduction

Learning from imbalanced data, where one or more classes are heavily under-represented, remains a central problem in machine learning. **[chawla2002smote]** proposed the Synthetic Minority Over-sampling Technique (SMOTE), which creates synthetic minority examples by interpolating between a minority instance and its nearest minority neighbors instead of replicating existing examples. Sixteen years later, **[fernandez2018smote]** marked its 15-year anniversary with a review of the progress and challenges of SMOTE-based learning. [TODO: Motivate a 25-year retrospective: growth of the literature since 2018, deep generative alternatives, and the lack of a review focused on societal impact.]

[TODO: State the three research questions of this survey: what has been done, how it benefited society, and what comes next.]

[TODO: Summarize the contributions and give a roadmap of Sections 2 to 9.]

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Table 1. Representative SMOTE Variants

Method	Year	Category	Key Idea
SMOTE (Ref.)	2002	Original	Interpolates between minority nearest neighbors
[TODO: variant]			

2 Background: The Class Imbalance Problem and the Original SMOTE

[TODO: Define the class imbalance problem, the imbalance ratio, and why standard learners and accuracy-based evaluation are biased toward the majority class.]

[TODO: Describe the original SMOTE algorithm of **[chawla2002smote]**: k-nearest minority neighbors, interpolation, the amount of over-sampling, and its combination with under-sampling. Consider an algorithm box and a small illustrative figure.]

[TODO: Summarize evaluation metrics used in the imbalanced-learning literature (e.g., ROC AUC, precision-recall, F-measure, G-mean, MCC).]

3 Review Methodology

This section describes how the literature reviewed in this survey was identified, screened, and analyzed. [TODO: One-paragraph overview of the review protocol.]

3.1 Search Strategy

[TODO: Databases and search engines, search strings, time window (2002 to present), and date of the last search.]

3.2 Inclusion and Exclusion Criteria

[TODO: Criteria for including a study, and reasons for exclusion.]

3.3 Study Selection and Data Extraction

[TODO: PRISMA-style flow (records identified, screened, assessed, included) as a figure, and the data extraction sheet. The literature is organized into survey, binary, multiclass, and both settings.]

4 Taxonomy of SMOTE Variants

[TODO: Introduce the taxonomy and the criteria used to build it. Planned categories: SMOTE variants, hybrid sampling, cleaning-based, cluster- and density-based, kernel and manifold, and comparisons with deep generative models (GAN, VAE, diffusion). Representative methods are listed in Table 1.]

5 SMOTE for Multiclass Imbalance

[TODO: Why multiclass imbalance differs from the binary case (multi-majority and multi-minority configurations, overlapping classes), decomposition strategies (one-vs-one, one-vs-all), and SMOTE variants designed directly for multiclass problems.]

6 Applications and Societal Benefits

[TODO: Domains where SMOTE-based methods were deployed or evaluated, and the measurable benefits reported: healthcare and medical diagnosis, fraud detection, cybersecurity, manufacturing fault detection, and environmental and social sciences.]

7 Empirical Comparisons and Benchmarks

A method in the SMOTE family is judged through an evaluation design: the data it is tested on, the learners trained on its output, the competitors it is compared with, the measures, the protocol that separates fitting from testing, and the statistics used to compare results. This section analyses those designs across the 71 empirical studies in our corpus. Every study is listed with its coded design in the evaluation map of Appendix B (Table 10).

Each study reports that its proposal works, but the evidence is rarely read across studies, and the designs differ enough that their numbers cannot be pooled. The analysis below supports one thesis:

> *Evaluation in this literature has broadened over 25 years, but the steps that most often decide a comparison (how settings are chosen, how measures are defined and how baselines are run) have not become more rigorous.*

We organise the evidence around six questions:

- **RQ1.** On which data are methods compared, and why those data? (Section 7.2)
- **RQ2.** Against which learners and competitors, and why those learners? (Section 7.3)
- **RQ3.** Which measures are reported, are they defined consistently, and is anything measured without a classifier? (Section 7.5)
- **RQ4.** How are training and test data separated, and how, and by which measure, are settings chosen? (Section 7.6)
- **RQ5.** How is a difference shown to be significant, and how are results reported? (Section 7.7)
- **RQ6.** What do the comparisons conclude, and do the conclusions agree across studies? (Section 7.9)

For each question we also ask whether studies that follow a practice are cited more often than studies of the same age that do not; Section 7.1 explains how.

Section 7.1 describes the corpus and the coding. Section 7.2 to Section 7.9 answer the six questions in turn, with Section 7.4 fixing the notation just before the measures are examined; Section 7.10 states the limits of this analysis, and Section 7.11 turns the findings into a minimum evaluation standard.

7.1 Corpus and Coding Scheme

The corpus is the SMOTE25 collection selected for this survey ([TODO: point to the selection procedure in Section 3]): 76 papers after duplicate copies were merged. Of these, 71 report an empirical evaluation on imbalanced data; the other 5 are two surveys (Ref.) and three foundational works that predate imbalance-aware evaluation (Ref.). Among the empirical studies, 39 evaluate binary problems, 19 multi-class problems and 13 both, and their number has grown with each period (Figure 1). Percentages in this section refer to the 71 empirical studies.

For each study we coded, from the full text and supplementary files, the benchmark data, base learners, competing methods, measures with their exact definitions, the validation protocol, how hyperparameters were chosen, the statistical tests, and the form in which results are reported. A field the paper does not state was coded as not reported and never inferred. Four derived codes involve judgement: the protocol class, the hyperparameter-selection class, the flags for measure-definition problems and the outcome class of Section 7.9. Sixteen of the most consequential codings were re-checked against the PDFs in a second pass, and all sixteen were confirmed.

For this version, a second pass over the 71 empirical studies recorded why each study made its choices: the reasons it gives for its data, base learners and measures; the measure and the data used to choose its settings; the methodological sources it cites for its tests; and any evidence about the resampled data that does not come from the score of a classifier. Every code from this pass carries a page reference and a verbatim quote of at most 30 words, so it can be checked against the paper. Where the two passes disagreed on how settings were chosen, the page-referenced reading was kept (Section 7.6). The second-pass codes have not yet been checked by a second reader [TODO: a co-author to verify the pass-2 quotes against the cited pages before submission].

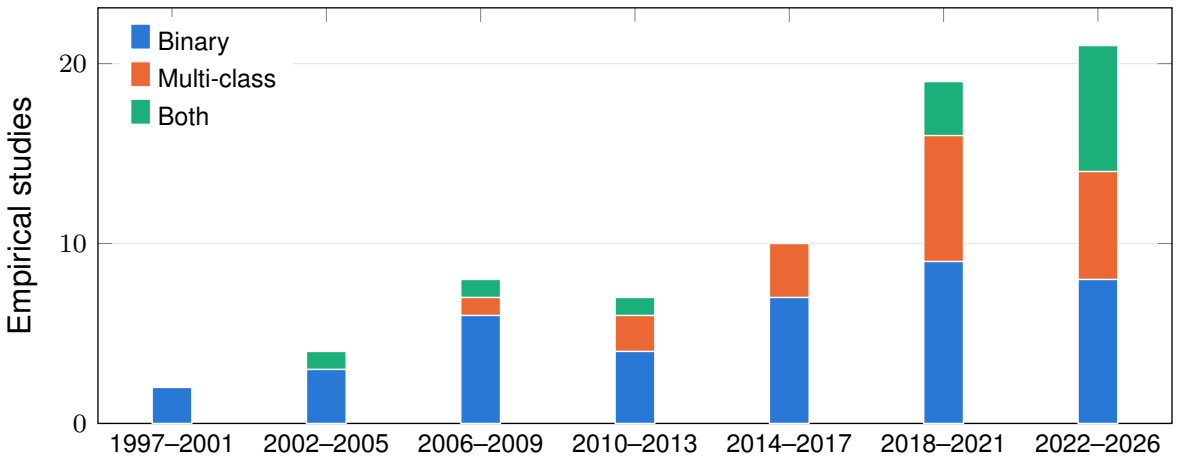


Fig. 1. Empirical Studies in the Corpus by Publication Period and Problem Setting

To see whether careful evaluation is rewarded, we took citation counts from OpenAlex on 18 October 2026; 75 of the 76 papers have a record. Older papers have had longer to be cited, and in this corpus each additional year of age multiplies the expected count by 1.19. We therefore compare practices with a regression of log(1 + citations) on publication year and an indicator of the practice, fitted to the 62 empirical studies published up to 2024 that have a record; the 8 studies from 2025 and 2026 are too recent for their counts to mean much. The exponentiated coefficient is a citation ratio: studies that follow a practice are cited that many times as often as studies of the same age that do not (Figure 2). The ratios describe association only. Venue, the novelty of the idea and the authors also drive citations and are not controlled, and with 24 comparisons about one interval would exclude 1 by chance.

The most cited studies are not the most carefully evaluated. Of the ten most cited empirical studies, 7 were published before 2010; they used a median of 8 real datasets, 4 ran a significance test and 1 released code. The other 52 studies used a median of 19.5 datasets and tested in 40 of 52 (77%). Once age is taken into account, one practice has an interval that excludes 1: studies with a measure-definition problem are cited about a third as often as others (0.33 (95% CI 0.14 to 0.74)). Releasing code (2.01 (95% CI 0.98 to 4.14)) and including SMOTE as a competitor (1.89 (95% CI 0.96 to 3.72)) come close. No practice of protocol, tuning or testing is associated with citations, so a study gains no visible credit for the steps that this section most often finds missing.

7.2 Benchmark Data (RQ1)

Benchmark data are the datasets on which a method is fitted and scored, together with the imbalance they carry. Evaluation relies almost entirely on public tabular benchmarks (Table 2). UCI datasets are used in 53 of 71 studies (75%) and the KEEL repository in 33 of 71 studies (46%); most KEEL imbalanced sets are binarised versions of UCI data [TODO: confirm this origin in the KEEL data-set paper (Alcalá-Fdez et al. 2011, cited below) and cite it here, or drop the clause]. 29 of 71 studies (41%) add synthetic data, mainly to show where a sampler places new points near the class boundary. Only 6 studies go beyond tabular data, to images, time series, text or mixed data (Ref.).

The number of real datasets per study has a median of 18 (interquartile range 9 to 30; range 1 to 86) and has grown steadily, from a median of 6 before 2010 to 21.5 since 2021 (Figure 3). Early method papers were often

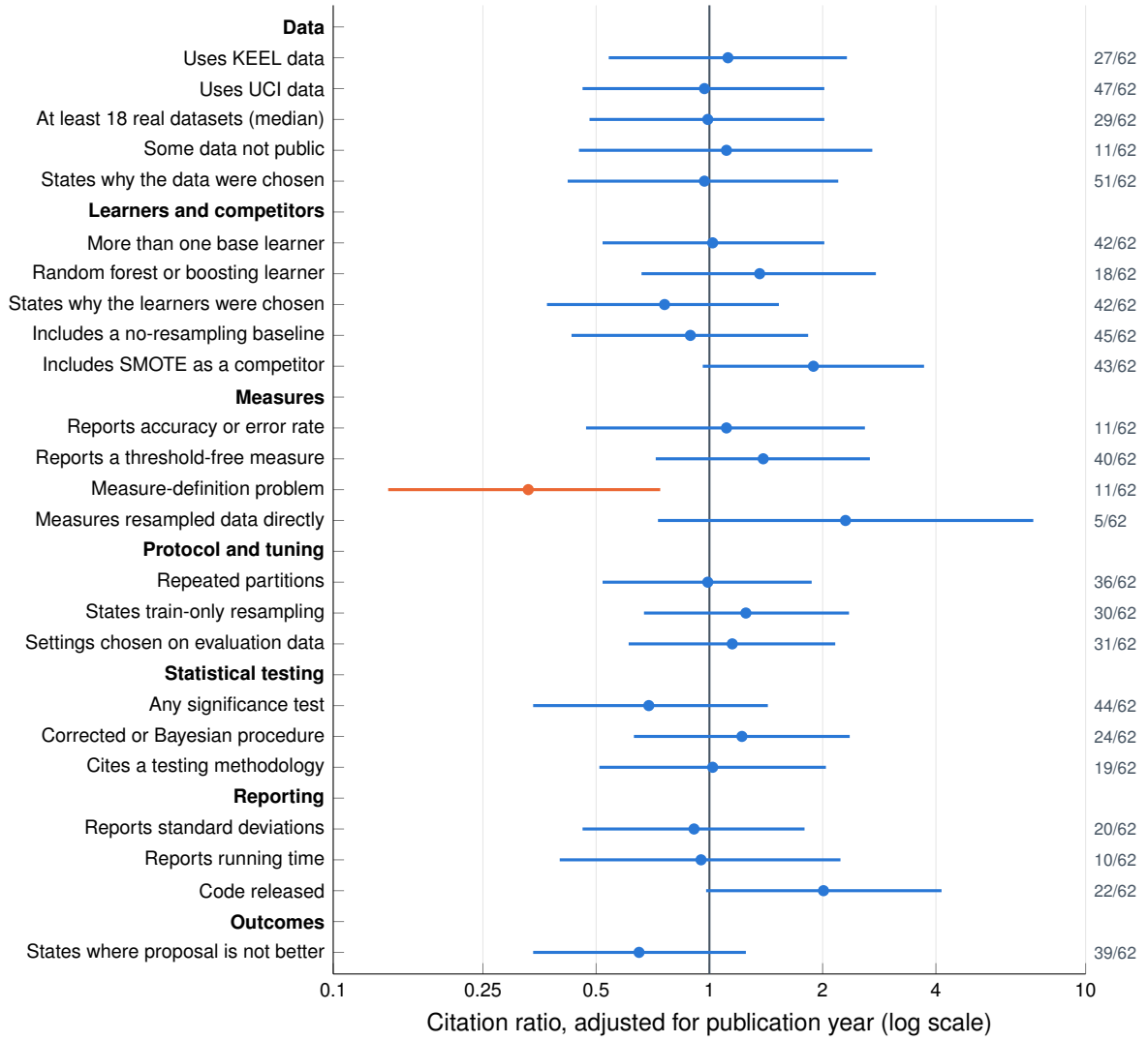


Fig. 2. Citation Ratio of Studies That Follow Each Evaluation Practice, Adjusted for Publication Year

shown on two to four datasets (Ref.), while recent large comparisons use 50 to 90 (Ref.). 58 of 71 studies (82%) report the imbalance ratio (IR) of their benchmarks.

Most studies say why they chose their data (58 of 71 studies (82%)), and the reasons concern coverage more than relevance to a task (Table 3): a range of imbalance ratios, sizes or dimensions (30 studies), a target level of imbalance (19), the set-up of a named earlier study (18), known difficulty factors such as overlap, noise or small disjuncts (17), comparability with published results (14) and public availability (12). Reuse is how particular collections spread. The multi-class KEEL sets selected in an earlier study of instance-level difficulty [TODO: cite Sáez, Krawczyk and Woźniak (2016)] were taken over by MC-RBO and MC-CCR and then by NROMM (Ref.);

Table 2. Sources of Benchmark Data

Source	Studies	Share	Typical use
UCI repository	53	75%	Original source of most tabular benchmarks
KEEL repository	33	46%	Pre-binarised imbalanced sets with fixed folds
Synthetic data	29	41%	Controlled overlap, noise or class shapes
Private or application data	15	21%	Case studies in one domain
Image benchmarks	6	8%	Imbalance induced in balanced image sets
Other repositories	13	18%	Kaggle, OpenML, imbalanced-learn, LIBSVM, UCR, PROMISE

A study can draw on several sources.

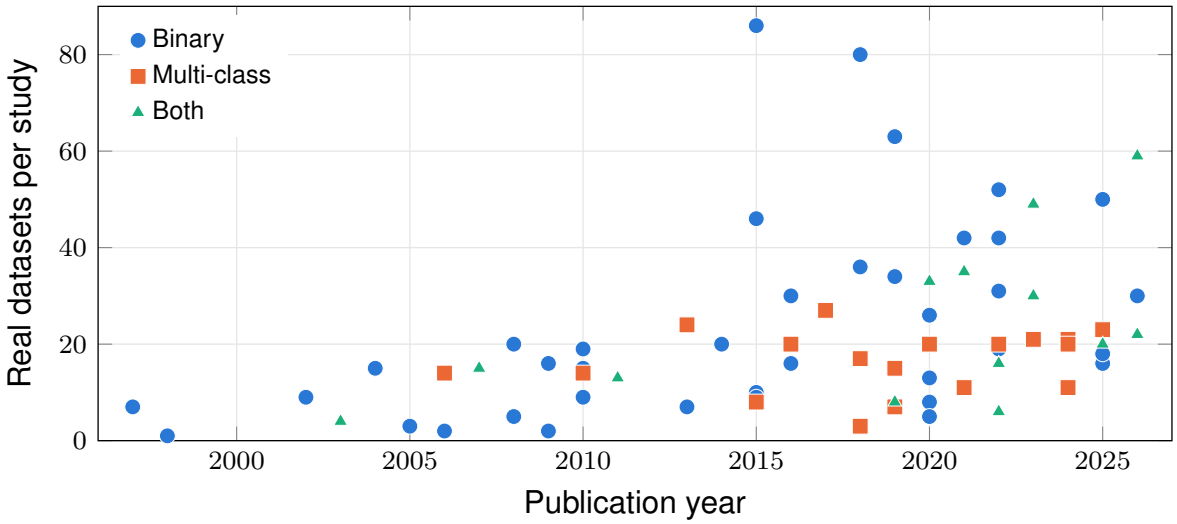


Fig. 3. Number of Real Benchmark Datasets per Study, by Publication Year and Problem Setting

EMDO reuses the AMDO sets in its first experiment (Ref.), and SMOTE-IPF takes its real datasets from the SPIDER2 study of noisy and borderline examples (Ref.).

Being public is not the same as being reproducible. The data are entirely public in 53 of 71 studies (75%), a table gives the source of each dataset in 42 studies, and links or repository identifiers appear in 42. However, 52 studies changed their data by binarising classes, merging them, injecting noise or subsampling, and only 10 of these released the changed versions. The others cannot be re-run on the same data even when the original repository is named.

The repositories themselves are among the most cited works in the field. The KEEL software and data-set papers have 1,505 and 2,337 citations in OpenAlex (Ref.), and the main record of the UCI repository has 23,780 (Ref.), more than any method in this corpus except SMOTE (32,518). The credit stays with the repositories. Studies that use KEEL data are cited about as often as studies of the same age that do not (1.12 (95% CI 0.54 to 2.32)), and the same holds for UCI data (0.97 (95% CI 0.46 to 2.02)), for using at least the median number of datasets (0.99 (95% CI 0.48 to 2.02)) and for stating why the data were chosen (0.97 (95% CI 0.42 to 2.20)).

Table 3. Reasons Given for Evaluation Choices

Choice	Reason given	Reasons given (studies)	Examples
Benchmark data	58 of 71	range of IR, size or dimension (30); a target imbalance level (19); set-up of a named earlier study (18); overlap, noise or small disjuncts (17); comparability with published results (14); public availability (12); an application domain (11)	(Ref.)
Base learners	49 of 71	widely used or well known (22); different learning paradigms (15); set-up of a named earlier study (11); simple, fast or interpretable (10); same learner for all methods (8); sensitive to imbalance (4); matches how the method works (4)	(Ref.)
Measures	70 of 71	standard in the literature (40); accuracy misleads under imbalance (39); focus on the minority class (25); independent of the threshold (18); suits several classes (18); costs of an application (7)	(Ref.)
Measure used to select settings	33 of 54	G-mean (15); AUC (12); F1 (8); AvAcc (4); sensitivity (recall) (4); MAUC (3)	(Ref.)

A study can give several reasons. Selection criterion: studies that select a parameter value, variant or resampling amount.

7.3 Base Learners and Competing Methods (RQ2)

A base learner is the classifier trained on the resampled data, and a resampler is judged only through it. C4.5/C5.0 trees are the most common learner (33 of 71 studies, 46%), followed by k-NN (28 of 71 studies, 39%), SVM (28 of 71 studies, 39%), CART (17 of 71 studies, 24%) and multilayer perceptrons (17 of 71 studies, 24%) (Figure 4). Random forests and gradient boosting appear in 2 of 27 studies before 2016 but in 33% (8 of 24) since 2021, while the use of C4.5 has declined. The median study uses 3 learners, yet 25 of 71 studies (35%) rely on a single one. That share has not fallen and stands at 38% (9 of 24) since 2021. With one learner, the effect of a sampler cannot be separated from its interaction with that learner. A recent pre-registered comparison found that a new oversampler was not significantly better than class weighting, random oversampling or no resampling across four learners that include random forest and gradient boosting (Ref.).

A competing method is any alternative run on the same benchmarks. Studies compare against a median of 7 competitors. SMOTE itself is a competitor in 50 of 71 studies (70%), and an untreated reference without resampling is included in 52 of 71 studies (73%). The remaining 19 studies report only gains of one sampler over another, so a reader cannot tell whether resampling helps at all on those benchmarks. In 6 studies some or all competitor results were copied from earlier publications instead of being re-run on the same folds (Ref.). In the most extreme case only the proposed method was run, and every competitor score was taken from a published benchmark (Ref.).

Studies say less about their learners than about their data: 49 of 71 studies (69%) give a reason (Table 3). The usual reason is that a learner is widely used or well known (22 studies), followed by covering different learning paradigms (15), following a named earlier study (11), simplicity (10) and using the same learner for every method (8). Only 4 studies choose a learner because it is sensitive to class imbalance, the property that makes it a meaningful test of a resampler (Ref.), and 4 because it fits how the method works. Following earlier work keeps C4.5 in use: SPE keeps it for a fair comparison with earlier ensembles, EMDO because its results are deterministic for fixed settings, and NROMM adopts the learners, folds and measures of MC-CCR and MC-RBO (Ref.).

Learner choices are not associated with citations either. Using more than one learner (1.02 (95% CI 0.52 to 2.02)), a random forest or boosting learner (1.36 (95% CI 0.66 to 2.77)) or an untreated baseline (0.89 (95% CI 0.43 to 1.83)) all give intervals that include 1. Studies that include SMOTE as a competitor are cited more (1.89 (95% CI 0.96 to 3.72)), but that interval also reaches 1.

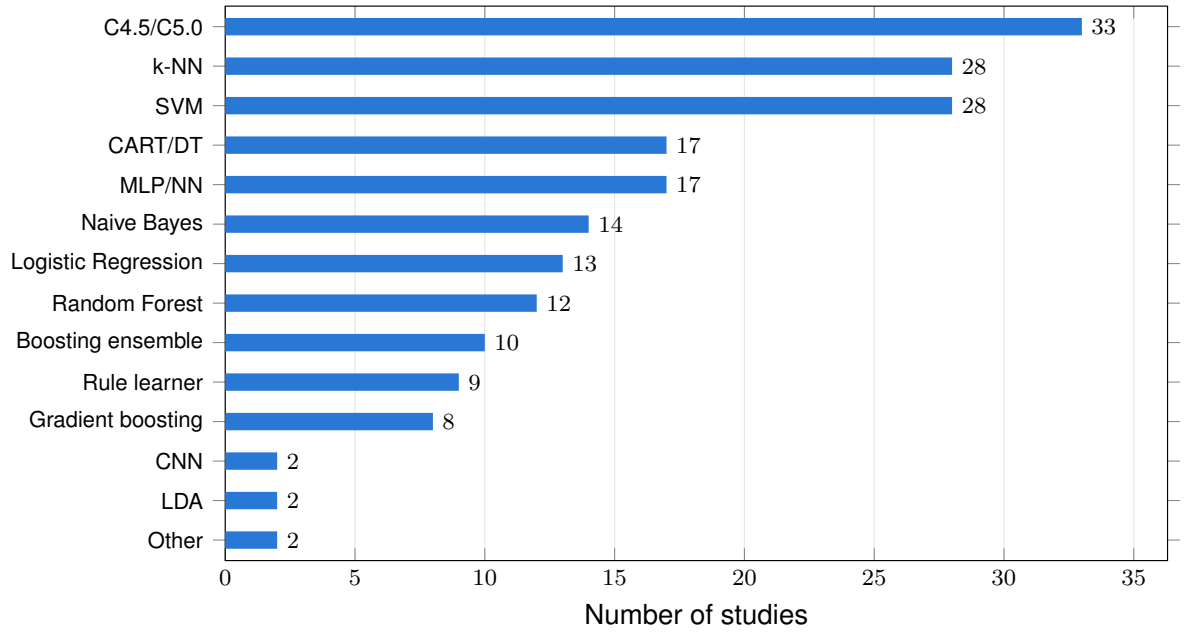


Fig. 4. Base Learner Families Used in the Empirical Studies

7.4 Notation and Measures

A measure summarises a confusion matrix, or a ranking of scores, into one number. For a binary problem we treat the minority class as positive and write TP, FN, FP and TN for the cells of the confusion matrix. The single-class rates are

$$\mathrm{TPR} = \frac{TP}{TP + FN}, \qquad \mathrm{TNR} = \frac{TN}{TN + FP}, \qquad \mathrm{Prec} = \frac{TP}{TP + FP} \tag{1}$$

where TPR is also called recall or sensitivity and TNR specificity. Measures that combine these rates at one decision threshold are

$$F_\beta = \frac{(1 + \beta^2)\,\mathrm{Prec} \cdot \mathrm{TPR}}{\beta^2\,\mathrm{Prec} + \mathrm{TPR}} \tag{2}$$

$$\text{G-mean} = \sqrt{\mathrm{TPR} \cdot \mathrm{TNR}}, \qquad \mathrm{BA} = \frac{\mathrm{TPR} + \mathrm{TNR}}{2} \tag{3}$$

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{4}$$

where β weights recall against precision ($\beta = 1$ gives F1) and BA is balanced accuracy. The area under the ROC curve does not depend on a threshold. For a scoring function s it is the probability that a random positive example is ranked above a random negative one,

$$\mathrm{AUC} = \Pr\big(s(x^+) > s(x^-)\big) \tag{5}$$

Table 4. Families of Performance Measures

Family	Measures	Advantages	Limitations	Studies	Examples
Overall	Accuracy, error rate	Simple; comparable with older work	Dominated by the majority class	13	(Ref.)
Single-class rates	Recall (TPR), specificity (TNR), precision	Interpretable per class	Trade off against each other; one rate alone rewards over-predicting the minority	32	(Ref.)
Threshold composites	F1, Fβ, G-mean, BA, MCC, kappa	One class-sensitive number	Depend on the decision threshold; F ignores true negatives	56	(Ref.)
Threshold-free ranking	ROC-AUC, PR-AUC, ROC curves	Independent of the threshold	ROC-AUC can look favourable under severe imbalance; PR-AUC depends on prevalence	46	(Ref.)
Multi-class aggregates	MG, AvAcc, MAUC, macro-F1, CBA, CEN	Weights every class equally	The averaging rule changes the result; MG is zero if one class has zero recall	21	(Ref.)

where ties count one half. For C classes, let R_i be the recall of class i and m_{ij} the number of class-i examples predicted as class j. The multi-class measures used in the corpus are

$$\mathrm{MG} = \Big(\prod_{i=1}^{C} R_i\Big)^{1/C}, \qquad \mathrm{AvAcc} = \frac{1}{C}\sum_{i=1}^{C} R_i \tag{6}$$

$$\mathrm{MAUC} = \frac{2}{C(C-1)} \sum_{i<j} \frac{A(i \mid j) + A(j \mid i)}{2} \tag{7}$$

$$\mathrm{CBA} = \frac{1}{C}\sum_{i=1}^{C} \frac{m_{ii}}{\max\big(\sum_j m_{ij},\ \sum_j m_{ji}\big)} \tag{8}$$

where MG is the class-wise geometric mean, AvAcc the average class accuracy, $A(i \mid j)$ the AUC of separating class i from class j using the scores for class i, and CBA the class balance accuracy. MAUC follows Hand and Till (Ref.) [TODO: cite the original sources of CBA (Mosley 2013) and confusion entropy (Wei et al. 2010)]. Table 4 groups the measures into families and lists what each family captures [TODO: cite methodological sources for the advantages and limitations listed in this table].

7.5 Performance Measures in Use (RQ3)

Overall accuracy or error rate appears in only 13 of 71 studies (18%), and never as the only measure. G-mean is the most used measure (41 of 71 studies, 58%), followed by AUC (33 of 71 studies, 46%), recall (29 of 71 studies, 41%), F1 (24 of 71 studies, 34%) and precision (20 of 71 studies, 28%); a typical study reports 4 measures (Figure 5). Binary and multi-class studies use different vocabularies. Binary studies centre on AUC, G-mean, and minority-class recall and F1. Multi-class studies use MG, MAUC (12 of 71 studies, 17%), AvAcc and, in a group of recent papers, CBA and confusion entropy (Ref.).

Threshold-free measures (ROC-AUC, MAUC or PR-AUC) appear in 46 of 71 studies (65%). The other 25 studies judge samplers only at the default decision threshold of the classifier, which mixes the quality of the synthetic data with the calibration of the classifier. Precision-recall summaries are rare, although they are more informative than ROC curves when positives are rare (Ref.): PR-AUC is reported in 5 studies and MCC in 4, nearly all published since 2018 (Ref.).

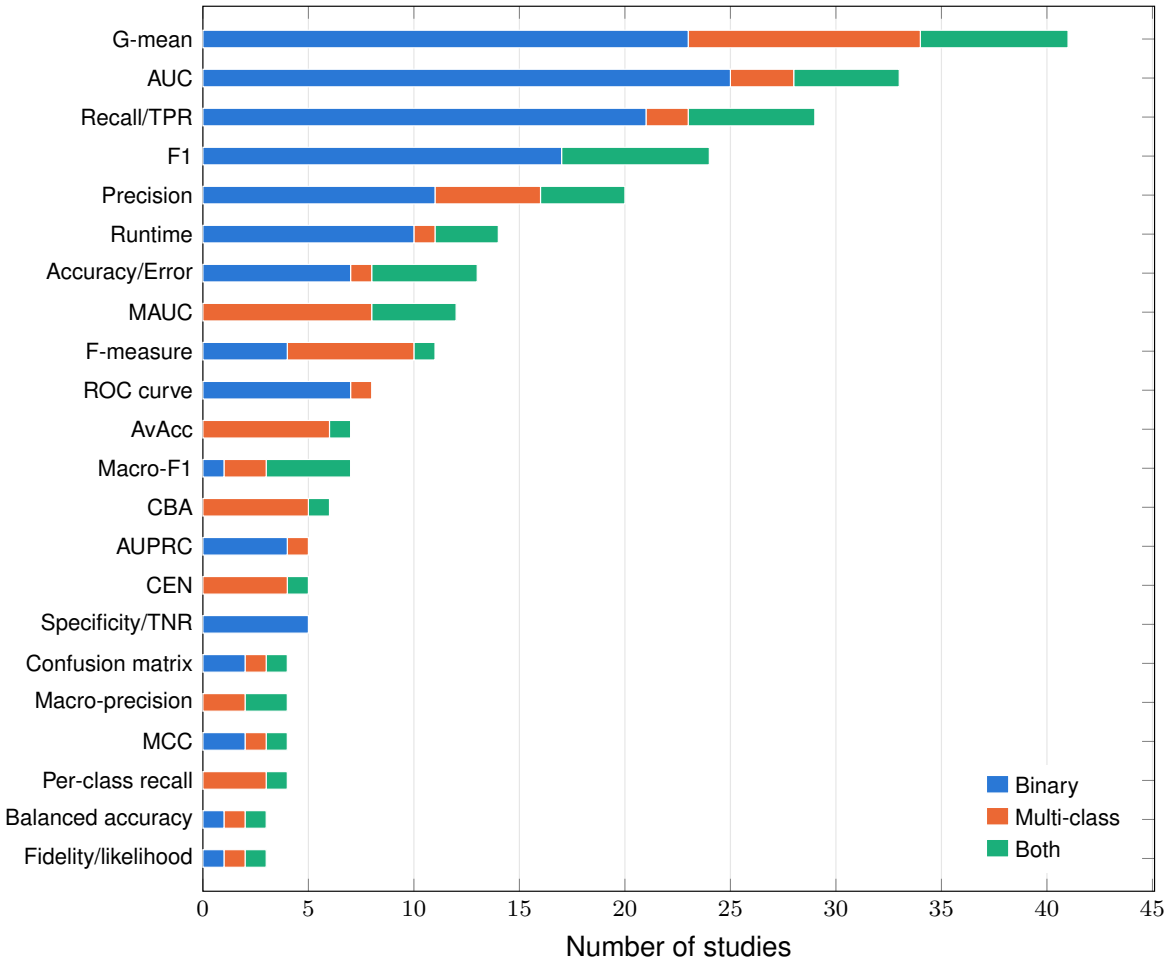


Fig. 5. Measures Used in at Least Three Empirical Studies, Split by Problem Setting

The same name does not always denote the quantity defined in Section 7.4. We found definition problems in 14 studies (Table 5). Two report as AUC the single-threshold value (TPR + TNR)/2, which is BA in Equation (3) and not the area of Equation (5) (Ref.). Two compute MAUC from hard-label confusion matrices instead of scores (Ref.). Two define AvAcc as the mean one-versus-rest accuracy, which true negatives inflate (Ref.). G-mean is sometimes the geometric mean of recall and precision instead of TPR and TNR (Ref.), and some studies never state how binary measures are averaged over classes (Ref.). Values that share a name are therefore not comparable across studies.

Nearly every study justifies its measures (70 of 71 studies (99%)), with the standard arguments (Table 3): accuracy misleads under imbalance (39 studies), the measure is standard in imbalanced learning (40), it focuses on the minority class (25), it does not depend on the threshold (18) or it suits several classes (18). These reasons explain why accuracy fell out of use, but not why one imbalance-aware measure is preferred to another, and only 7

Table 5. Measure-Definition Problems Found in the Corpus

Study	Setting	Problem
SCUT (Ref.)	multiclass	multi-class averaging of G-mean, F-measure and AUC not defined
CGMOS (Ref.)	binary	'G-score' = sqrt(Precision x Recall), reported for both classes as positive
AMDO (Ref.)	multiclass	Pavg described as precision but equated with average accuracy
DLexiBoost (Ref.)	both	'AUC' computed from crisp predictions as (1+TPR-FPR)/2 (= balanced accuracy)
SPE (Ref.)	binary	G-mean defined as sqrt(Recall x Precision) instead of sqrt(TPR x TNR)
EMDO (Ref.)	multiclass	per-class precision called 'accuracy' (Pmin/Pavg)
GDO (Ref.)	binary	'AUC' defined as (Sens+Spec)/2, i.e. single-threshold balanced accuracy
SOS (Ref.)	both	weighted-F1 (majority-dominated) as the sole main metric
SHSampler (Ref.)	multiclass	MAUC computed from hard-label confusion matrices, not scores
CBHSID (Ref.)	both	multi-class averaging of Recall/F1 not stated; F1 misdefined
MC-MBRC (Ref.)	multiclass	AvAcc defined as mean one-vs-rest accuracy (inflated by true negatives)
GDHS (Ref.)	both	MAUC computed from hard-label confusion matrices, not scores
MC-NCLWO (Ref.)	multiclass	AvAcc defined as mean one-vs-rest accuracy (inflated by true negatives)
Loyalty-SMOTE (Ref.)	both	implausible AUC values (e.g. 0) and averages inconsistent with tables

Table 6. Evidence Other Than Classifier Scores

Evidence	What it shows	Studies	Examples
Visual inspection	2-D plots of where synthetic points fall; PCA or t-SNE projections; histograms of real and synthetic columns	33	(Ref.)
Analytical properties	Theorems or bounds on the sampler; big-O time or memory	31	(Ref.)
Model diagnostics without a score	Size of trees or rule sets learned from the resampled data; diversity of ensemble members	5	(Ref.)
Direct measurement of the resampled data	Distance or likelihood between real and synthetic distributions (MMD, FID, likelihood fitness); share of synthetic examples removed as noise; safe, borderline, rare and outlier composition	6	(Ref.)
Classifier scores only	No evidence of any of the kinds above	23	(Ref.)

A study can provide several kinds of evidence; the counts overlap.

studies relate their measures to the costs of an application. Measure-definition problems are the one practice clearly associated with fewer citations (0.33 (95% CI 0.14 to 0.74)); reporting accuracy (1.11 (95% CI 0.47 to 2.59)) or a threshold-free measure (1.39 (95% CI 0.72 to 2.67)) shows no association.

Every study judges a resampler through the scores of a classifier, and none evaluates resampling without one. Other evidence is common but mostly visual or analytical (Table 6): 33 studies plot where synthetic points fall or project the resampled data, and 31 give theorems, bounds or big-O costs. Model diagnostics that do not use a score, such as the size of trees learned from the resampled data or the diversity of ensemble members, appear in 5 (Ref.). Direct measurement of the resampled data is rare (6 of 71 studies (8%)): distances or likelihoods between real and synthetic distributions (Ref.), the share of synthetic examples later removed as noise (Ref.), and the composition of safe, borderline and hard examples after resampling (Ref.). Such measures can show why a sampler helps or fails on a dataset, which a score cannot. The remaining 23 studies offer classifier scores only. The few studies that measure the data directly are cited more (2.30 (95% CI 0.73 to 7.26)), but only 5 of them are old enough to compare and the interval is wide.

Table 7. Validation Protocols

Protocol	Description	Advantages	Limitations	Studies	Examples
Repeated k-fold CV	k folds, reshuffled and repeated	Uses all data; variance can be estimated	Costly; folds overlap across repeats	33	(Ref.)
Single k-fold CV	One run of k folds	Uses all data once	One estimate per dataset; sensitive to the split	16	(Ref.)
5×2 CV	Five repeats of two folds	Suits paired tests on each dataset	Half the data for training	4	(Ref.)
Repeated hold-out	Several random train/test splits	Simple; variance can be estimated	Test sets overlap; few minority test examples	6	(Ref.)
Single hold-out	One train/test split	Cheap for large data	One estimate; high variance	8	(Ref.)
Fixed benchmark split	The split shipped with the dataset	Comparable with other work on that benchmark	One split; imbalance often induced artificially	3	(Ref.)

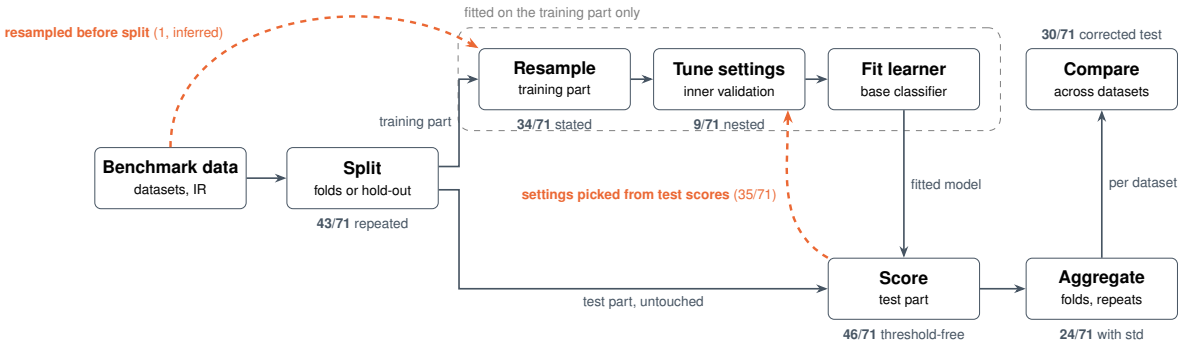


Fig. 6. Where Evaluation Decisions Sit in One Run of a Resampling Experiment

7.6 Validation Protocol and Hyperparameter Selection (RQ4)

A validation protocol decides which examples are used to fit the resampler and the learner and which are used to score them. Stratified k-fold cross-validation is the norm (Table 7; [TODO: cite a methodological source for the advantages and limitations of each protocol]): 33 of 71 studies (46%) repeat it and 16 of 71 studies (23%) run it once, with k = 10 (23 studies) and k = 5 (20) as the usual choices. Hold-out designs remain common, and 3 deep or generative studies use fixed benchmark splits (Ref.). One study does not describe its protocol at all (Ref.). Overall, 43 of 71 studies (61%) average results over repeated partitions; the rest report one estimate per dataset.

Figure 6 places the remaining decisions in one run of an experiment. Resampling and tuning belong inside the training part. Only 34 of 71 studies (48%) state or describe unambiguously that resampling was confined to training data, and 36 studies do not say. For one study the reported numbers of synthetic samples equal the full minority-class size, which implies that the whole dataset was oversampled before 5-fold CV (Ref.); another appears to normalise features before splitting (Ref.).

Hyperparameter selection is the weakest step (Figure 7). Only 9 of 71 studies (13%) tune parameters inside the training data, through nested CV or a disjoint set of tuning datasets (Ref.). 20 of 71 studies (28%) use fixed or default settings for every method, which is even-handed but can penalise methods with poor defaults. In 35 of 71 studies (49%), the reported configuration of the proposed method, or the amount of resampling, was chosen by

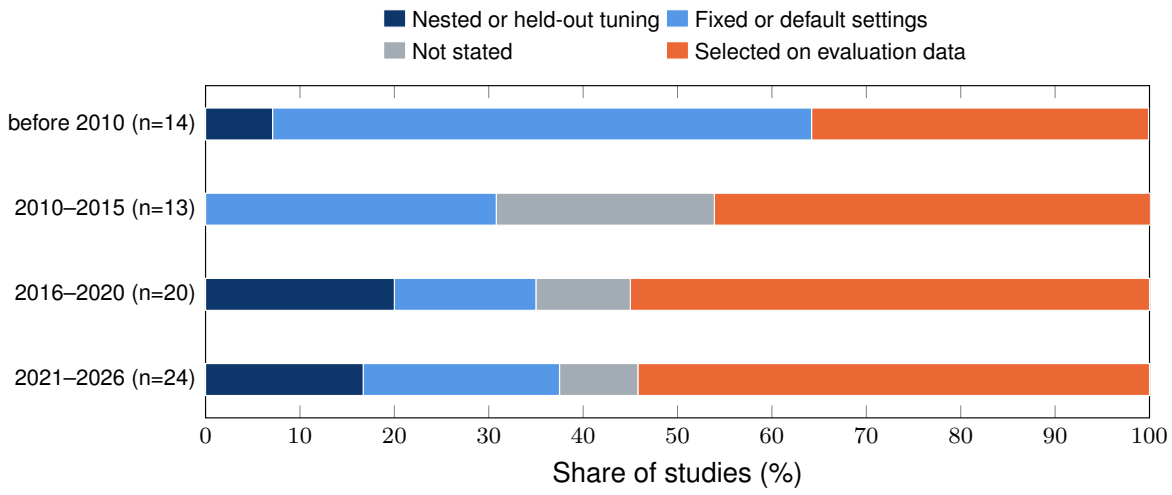


Fig. 7. How Hyperparameters Were Selected, by Period

looking at results on the evaluation data, typically by reporting the best point of a grid evaluated on the same folds; the remaining 7 do not say how settings were chosen. Search budgets are often unequal: G-SMOTE is selected from 294 configurations against 2 for SMOTE (Ref.), and k-means SMOTE keeps, for each measure, the best cross-validated score over its grid (Ref.). This share rose from 36% (5 of 14) before 2010 to 54% (13 of 24) since 2021, over the same period in which statistical testing became common (Section 7.7).

The page-referenced second pass separates two kinds of evidence for choosing settings on evaluation data. In 25 studies the paper says so, for example by keeping the best value of a grid run on the reported folds. In 10 more, values were selected after preliminary runs or enumerated to obtain the best result, and no separate validation data are described (Ref.). The counts above combine both; the first pass had coded 29. Choosing a model on the data that are then used to estimate its performance biases the estimate upwards (Ref.).

Which measure decides the setting? Of the 54 studies that select a parameter value, variant or resampling amount, 33 name the measure used, 4 use an internal criterion of the method such as a likelihood, and 17 do not say (Table 3). G-mean (15 studies) and AUC (12) are the usual criteria, followed by F1 (8) and AvAcc (4). Sensitivity alone decides the selection in only 2 studies, both inside an internal stopping or model-choice rule (Ref.), so selecting on recall is not the norm. In 33 of 33 studies that name a criterion, it is also one of the measures they report, so when selection uses the evaluation folds the proposal is tuned on exactly the number the reader sees.

Choosing settings on evaluation data carries no citation penalty (1.15 (95% CI 0.61 to 2.16)), and neither repeated partitions (0.99 (95% CI 0.52 to 1.87)) nor a statement of train-only resampling (1.25 (95% CI 0.67 to 2.35)) is associated with citations.

7.7 Statistical Testing (RQ5)

A significance test asks whether a difference in scores across datasets is larger than chance would produce. 52 of 71 studies (73%) use one. The Wilcoxon signed-rank test (26 studies) and the Friedman test (25) dominate, with Holm (10), Bonferroni–Dunn (6), Shaffer (4) and Nemenyi (4) as post-hoc procedures. Grouped by what they control (Figure 8): 27 of 71 studies (38%) apply a corrected multiple-comparison procedure, in 23 cases Friedman followed by a post-hoc test against the proposed method; 19 of 71 studies (27%) use only pairwise tests without

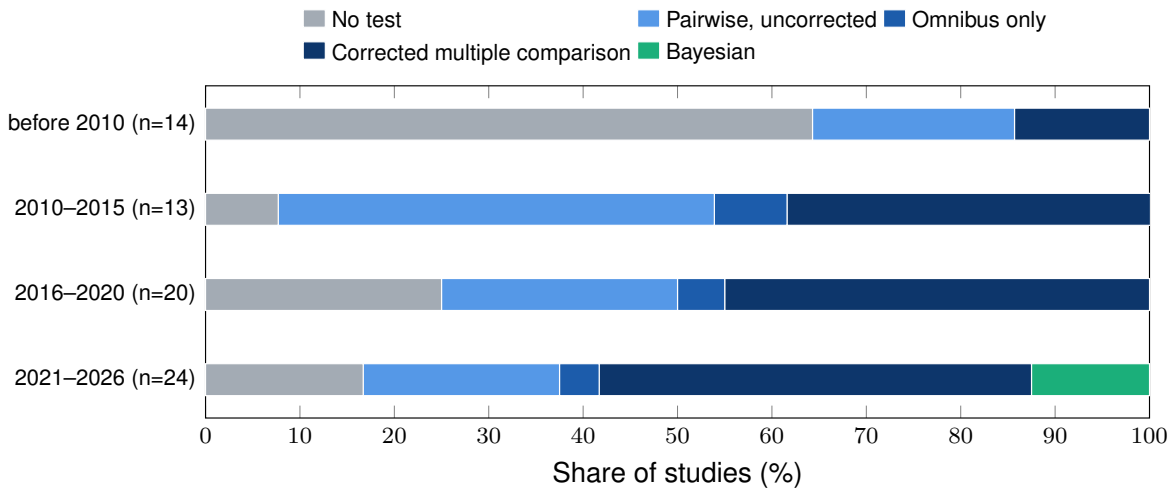


Fig. 8. Statistical Testing by Period

correction, although they compare the proposal with a median of 6 competitors; 3 stop at an omnibus test; 3 use Bayesian signed-rank tests (Ref.); and 19 of 71 studies (27%) report no test.

The share of studies with any test rose from 36% (5 of 14) before 2010 to 83% (20 of 24) since 2021, and the share with corrected or Bayesian procedures from 14% (2 of 14) to 58% (14 of 24). Inconsistencies remain, such as a Nemenyi diagram drawn after a non-significant Friedman test (Ref.), a Nemenyi test without an omnibus test (Ref.), and two different post-hoc procedures named for the same analysis (Ref.). Tests are also run on the folds used to pick hyperparameters (Section 7.6), so a significant result does not remove selection bias.

25 of 71 studies (35%) cite a methodological source for their tests. The most cited is Demšar's comparison of classifiers over multiple datasets (19 studies) (Ref.), which has 11,043 citations in OpenAlex. Later tutorials supply the Holm, Shaffer and aligned-rank procedures (Ref.), and the Bayesian tests follow Benavoli et al. (Ref.). Some cited sources do not describe the test that was run: one study cites the design of a variable-selection benchmark for its Friedman and Holm tests (Ref.), and another cites a guide to normality tests for its Wilcoxon test (Ref.). Testing is not associated with citations: the ratio is 0.69 (95% CI 0.34 to 1.43) for any test, 1.22 (95% CI 0.63 to 2.36) for a corrected or Bayesian procedure and 1.02 (95% CI 0.51 to 2.04) for citing a methodology.

7.8 Reporting of Results (RQ5)

Reporting is the form in which scores reach the reader. Results are mostly given as per-dataset tables (58 of 71 studies, 82%), usually with the best value in bold, and summarised by average ranks (38 of 71 studies, 54%) and tables of p-values (32 of 71 studies, 45%); win/tie/loss counts appear in 19 studies. Variability is less visible: only 24 of 71 studies (34%) give standard deviations with their means. Visual evidence is common, with 2-D plots of synthetic samples in 25 studies, parameter-sensitivity curves in 26 and ablation studies in 23. ROC curves (6), box plots (6) and critical-difference diagrams (4) are rare.

Cost and reproducibility receive less attention. Only 14 of 71 studies (20%) measure running time, although many proposals add clustering, optimisation or generative training on top of SMOTE. Code is available for 27 of 71 studies (38%), rising from none before 2010 to 65% (13 of 20) in 2016 to 2020 and 50% (12 of 24) since 2021

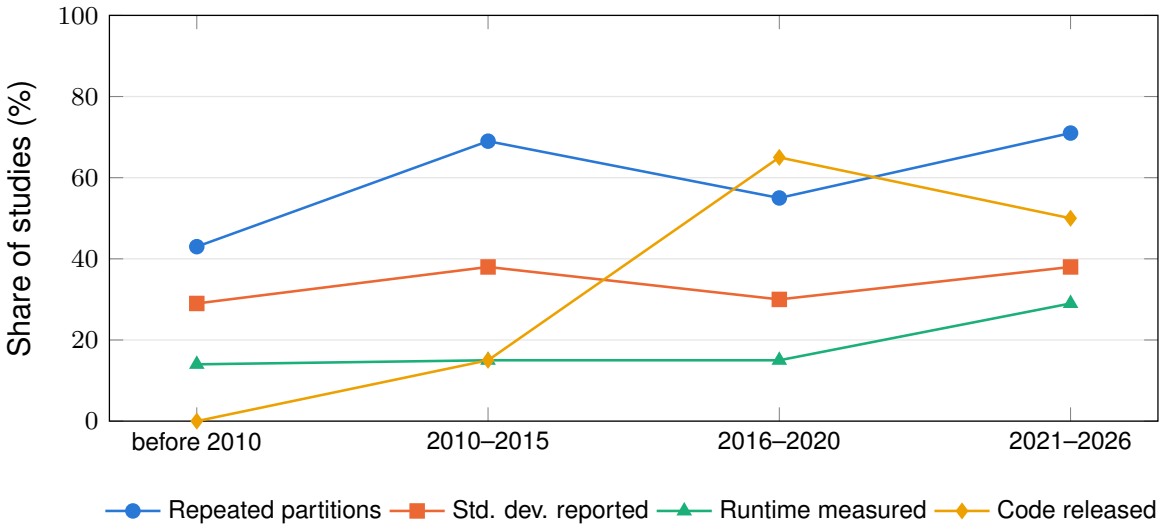


Fig. 9. Reporting Practices by Period

Table 8. Outcomes Reported by the Studies for Their Own Proposals

Outcome	Description	Studies	Examples
A. Best, no stated exception	The proposal is reported best on all summary measures and learners	27	(Ref.)
B. Best, with stated exceptions	Best overall, but the authors name a measure, learner, competitor or data condition where it is not better	41	(Ref.)
C. No advantage over simple baselines	Not better than no resampling, class weighting or random oversampling	1	(Ref.)
D. Comparative study	No new method; compares existing ones	2	(Ref.)

(Figure 9). 18 studies move per-dataset results to supplementary files, and some main texts show only ranks (Ref.) or only figures (Ref.), which prevents re-analysis. We also found tables that contradict the text of the same paper (Ref.).

Code release is the reporting practice most closely associated with citations: studies that release code are cited about twice as often as studies of the same age (2.01 (95% CI 0.98 to 4.14)), although the interval reaches 1. A released implementation also makes a method easier to include as a competitor in later studies. Reporting standard deviations (0.91 (95% CI 0.46 to 1.79)) or running time (0.95 (95% CI 0.40 to 2.23)) shows no association.

7.9 What the Comparisons Conclude (RQ6)

An outcome is what a study concludes about its own proposal. We coded it into four classes (Table 8). Of the 69 studies that propose a method, 68 report it as the best overall: 27 with no stated exception and 41 with an exception the authors state themselves, such as a measure, learner, competitor or data condition where the proposal is not better or not significantly better. Only one reports no advantage over simple baselines (Ref.), and that study is also the only one that pre-registered its protocol.

These wins do not add up to a ranking. A method that leads in one study is placed below the proposal in every later study that includes it: MDO is a competitor in 13 later studies and MC-CCR in 7, and each of those studies reports its own method ahead on its main summary measure (Ref.). Because each comparison uses its own folds, settings and measure definitions, the corpus cannot say which of these methods is better, only that each is better under the protocol of the paper that proposed it.

Modest conclusions are not penalised: studies that state where their proposal is not better are cited about as often as studies that claim an unqualified win (0.65 (95% CI 0.34 to 1.25)).

7.10 Threats to Validity

Six limits apply to the counts above. First, the corpus is the collection assembled for this survey, not an exhaustive sample, so shares describe these 71 studies and may not hold for the whole literature [TODO: align with the selection procedure in Section 3]. Second, coding was done by one extraction pass per paper; the four judgement codes were not double-coded, although the sixteen codings that the argument relies on most were re-checked against the PDFs. Third, some tables are images in the PDFs, so a few counts of datasets, classes or imbalance ratios are missing, and missing values were left empty rather than estimated. Fourth, each study is counted once per attribute, so a study that reports one measure in a footnote counts the same as one that builds its argument on it. Fifth, the reasons, selection criteria and evidence codes of the second pass come from one reading per paper; each carries a quote and a page, but a second reader has not yet checked them. Sixth, citation counts come from one index on one date. OpenAlex splits some papers over several records and indexes some only as preprints; we merged one split record and used the journal record where the corpus holds a dissertation chapter. Its counts are also lower than those of Google Scholar. The citation ratios are associations in a small corpus, not effects of the practices.

7.11 A Minimum Evaluation Standard

The analysis answers its six questions as follows. Methods are compared on more data and with more learners than before (RQ1, RQ2), but a third still use one learner and a quarter omit the untreated baseline. Measures share names without sharing definitions (RQ3). Settings are often chosen on the evaluation data, and this has become more common (RQ4), so the wider use of statistical tests (RQ5) does not make the reported gains safer. Nearly every study concludes that its proposal is best, and those conclusions do not form a consistent ranking (RQ6). Most studies say why they chose their data, fewer why they chose their learners, and settings are tuned and reported on the same measure. Influence does not follow rigour: the most cited studies used few datasets and seldom tested, and after adjusting for age no practice of protocol, tuning or testing is associated with citations. Table 9 pairs each observed problem with a remedy. We recommend that every new SMOTE-family method be evaluated to this standard.

A shared benchmark that fixes the folds, the tuning budget and the measure definitions, and that is re-run when a new method appears, would let these comparisons accumulate instead of restarting with each paper [TODO: link to the future-work agenda in Section 8].

8 Limitations, Open Challenges, and Future Directions

[TODO: Limitations of SMOTE-based methods (noise, overlap, high dimensionality, categorical and mixed features, data streams, big data), open challenges, and research gaps that motivate future over-sampling methods.]

9 Conclusion

[TODO: Answer the three research questions and close with the outlook for the next decade of SMOTE research.]

Table 9. Observed Problems and Recommended Practice

Observed problem	Studies	Recommended practice
One base learner	25	Use at least three learners from different families, including a strong ensemble such as random forest or gradient boosting.
No untreated baseline	19	Include no resampling, class weighting, random oversampling and SMOTE.
Reasons for learners not stated	22	Say why each learner was chosen, preferably for its sensitivity to class imbalance or to cover distinct learning paradigms.
Changed datasets not released	42	Release binarised, merged, subsampled or noise-injected versions together with their fold assignments.
Copied competitor results	6	Re-run every competitor on identical folds.
Threshold-dependent measures only	25	Report one threshold-free measure computed from scores and one class-balanced threshold measure.
Non-standard measure definitions	14	Give the formula of every measure and the multi-class averaging rule.
Train-only resampling not stated	37	State that resampling, scaling and imputation are fitted on training folds only.
Selection criterion not named	17	Name the measure and the data used to select every setting, and report results on data not used for selection.
Settings chosen on evaluation data	35	Tune every method with nested CV or a disjoint tuning suite under the same search budget.
Classifier scores only	65	Add one direct check of the resampled data, such as a distance between real and synthetic distributions or the share of synthetic examples that fall in majority regions.
No test, or uncorrected pairwise tests	38	Use the Friedman test with Holm or Shaffer post-hoc correction, or Bayesian tests, and report effect sizes.
No variability reported	47	Publish per-dataset means and standard deviations.
No runtime or no code	57 / 44	Report running time and release code and random seeds.

Studies: number of the 71 empirical studies showing the problem.

Acknowledgments

[TODO: Funding sources, grant numbers, and thanks to individuals and groups who assisted.]

References (stub)

A Reproducibility Checklist for JAIR

Select the answers that apply to your research – one per item.

All articles:

(1) All claims investigated in this work are clearly stated. [yes/partially/no]
(2) Clear explanations are given how the work reported substantiates the claims. [yes/partially/no]
(3) Limitations or technical assumptions are stated clearly and explicitly. [yes/partially/no]
(4) Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed. [yes/partially/no/NA]
(5) Motivation is provided for all design choices, including algorithms, implementation choices, parameters, data sets and experimental protocols beyond metrics. [yes/partially/no]

Articles containing theoretical contributions:

Does this paper make theoretical contributions? [yes/no]
If yes, please complete the list below.

(1) All assumptions and restrictions are stated clearly and formally. [yes/partially/no]
(2) All novel claims are stated formally (e.g., in theorem statements). [yes/partially/no]
(3) Proofs of all non-trivial claims are provided in sufficient detail to permit verification by readers with a reasonable degree of expertise (e.g., that expected from a PhD candidate in the same area of AI). [yes/partially/no]
(4) Complex formalism, such as definitions or proofs, is motivated and explained clearly. [yes/partially/no]
(5) The use of mathematical notation and formalism serves the purpose of enhancing clarity and precision; gratuitous use of mathematical formalism (i.e., use that does not enhance clarity or precision) is avoided. [yes/partially/no]
(6) Appropriate citations are given for all non-trivial theoretical tools and techniques. [yes/partially/no]

Articles reporting on computational experiments:

Does this paper include computational experiments? [yes/no]
If yes, please complete the list below.

(1) All source code required for conducting experiments is included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for source code readability and documentation as well as for long-term accessibility. [yes/partially/no]
(2) The source code comes with a license that allows free usage for reproducibility purposes. [yes/partially/no]
(3) The source code comes with a license that allows free usage for research purposes in general. [yes/partially/no]
(4) Raw, unaggregated data from all experiments is included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for long-term accessibility. [yes/partially/no]
(5) The unaggregated data comes with a license that allows free usage for reproducibility purposes. [yes/partially/no]
(6) The unaggregated data comes with a license that allows free usage for research purposes in general. [yes/partially/no]
(7) If an algorithm depends on randomness, then the method used for generating random numbers and for setting seeds is described in a way sufficient to allow replication of results. [yes/partially/no/NA]
(8) The execution environment for experiments, the computing infrastructure (hardware and software) used for running them, is described, including GPU/CPU makes and models; amount of memory (cache and RAM); make and version of operating system; names and versions of relevant software libraries and frameworks. [yes/partially/no]
(9) The evaluation metrics used in experiments are clearly explained and their choice is explicitly motivated. [yes/partially/no]
(10) The number of algorithm runs used to compute each result is reported. [yes/no]
(11) Reported results have not been "cherry-picked" by silently ignoring unsuccessful or unsatisfactory experiments. [yes/partially/no]
(12) Analysis of results goes beyond single-dimensional summaries of performance (e.g., average, median) to include measures of variation, confidence, or other distributional information. [yes/no]
(13) All (hyper-) parameter settings for the algorithms/methods used in experiments have been reported, along with the rationale or method for determining them. [yes/partially/no/NA]
(14) The number and range of (hyper-) parameter settings explored prior to conducting final experiments have been indicated, along with the effort spent on (hyper-) parameter optimisation. [yes/partially/no/NA]
(15) Appropriately chosen statistical hypothesis tests are used to establish statistical significance in the presence of noise effects. [yes/partially/no/NA]

Articles using data sets:

Does this work rely on one or more data sets (possibly obtained from a benchmark generator or similar software artifact)? [yes/no]

If yes, please complete the list below.

(1) All newly introduced data sets are included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for long-term accessibility with a license that allows free usage for research purposes. [yes/partially/no/NA]
(2) The newly introduced data set comes with a license that allows free usage for reproducibility purposes. [yes/partially/no]
(3) The newly introduced data set comes with a license that allows free usage for research purposes in general. [yes/partially/no]
(4) All data sets drawn from the literature or other public sources (potentially including authors' own previously published work) are accompanied by appropriate citations. [yes/no/NA]
(5) All data sets drawn from the existing literature (potentially including authors' own previously published work) are publicly available. [yes/partially/no/NA]
(6) All new data sets and data sets that are not publicly available are described in detail, including relevant statistics, the data collection process and annotation process if relevant. [yes/partially/no/NA]
(7) All methods used for preprocessing, augmenting, batching or splitting data sets (e.g., in the context of hold-out or cross-validation) are described in detail. [yes/partially/no/NA]

Explanations on any of the answers above (optional):

[Text here; please keep this brief.]

B Evaluation Map of the Empirical Studies

Table 10 lists the coded evaluation design of each of the 71 empirical studies analysed in Section 7, grouped by problem setting and ordered by year.

Table 10. Evaluation Map of the Empirical Studies

Study	Year	Set.	DS	Classes	Base learners	Measures	Protocol	Test	Out.
OSS (Ref.)	1997	B	7	2	k-NN, C4.5/C5.0	G-mean, Recall/TPR, Specificity/TNR, Accuracy/Error	k-CV	none	B
SHRINK (Ref.)	1998	B	1	2	C4.5/C5.0, k-NN	G-mean, Recall/TPR, Specificity/TNR, ROC curve	rep. 10-CV	t-test	B
SMOTE (Ref.)	2002	B	9	2	C4.5/C5.0, Rule learner, Naive Bayes	AUC, ROC curve, Recall/TPR, Specificity/TNR	10-CV	none	A
Batista et al. (Ref.)	2004	B	15	2	C4.5/C5.0	AUC, ROC curve	10-CV	Hsu MCB	B
Borderline-SMOTE (Ref.)	2005	B	3	2	C4.5/C5.0	Recall/TPR, F1	rep. 10-CV	none	B
Cluster-SMOTE (Ref.)	2006	B	2	2	Rule learner	ROC curve, Recall/TPR, FPR	hold-out	none	B
ADASYN (Ref.)	2008	B	5	2	CART/DT	Accuracy/Error, Precision, Recall/TPR, F1, G-mean	rep. hold-out	none	B
LS (Ref.)	2008	B	20	2	C4.5/C5.0, SVM	AUC	5×2 CV	CV F-test, Friedman, Holm	A
EasyEnsemble (Ref.)	2009	B	16	2	Boosting ensemble, Random Forest	AUC, F1, G-mean, Runtime	rep. 10-CV	t-test	B
Safe-Level-SMOTE (Ref.)	2009	B	2	2	C4.5/C5.0, Naive Bayes, SVM	Precision, Recall/TPR, F1, AUC	10-CV	none	B
SBC (Ref.)	2009	B	2	2	MLP/NN	Precision, Recall/TPR, F1, Runtime	hold-out	none	A

Study	Year	Set.	DS	Classes	Base learners	Measures	Protocol	Test	Out.
RAMOBoost (Ref.)	2010	B	19	2	MLP/NN	Accuracy/Error, Precision, Recall/TPR, F-measure, G-mean, ROC curve, AUC, Runtime	rep. hold-out	Wilcoxon signed-rank	B
RUSBoost (Ref.)	2010	B	15	2	C4.5/C5.0, Rule learner, Naive Bayes	AUC, AUPRC, F1	rep. 10-CV	ANOVA, Tukey HSD, t-test	B
SPIDER2 (Ref.)	2010	B	9	2	Rule learner, C4.5/C5.0	Recall/TPR, Specificity/TNR, G-mean, Accuracy/Error	rep. 10-CV	Wilcoxon signed-rank	B
INOS (Ref.)	2013	B	7	2	SVM	F1, G-mean, Precision, Recall/TPR, Specificity/TNR, Runtime	rep. hold-out	none	B
MWMOTE (Ref.)	2014	B	20	2	MLP/NN, Boosting ensemble, k-NN, C4.5/C5.0	Accuracy/Error, Precision, Recall/TPR, F1, G-mean, AUC, ROC curve	10-CV	Wilcoxon signed-rank	B
RACOG (Ref.)	2015	B	10	2	C4.5/C5.0, SVM, k-NN, Logistic Regression	Recall/TPR, G-mean, AUC	5-CV	t-test	B
Random Balance (Ref.)	2015	B	86	2	C4.5/C5.0, k-NN, SVM	AUC, F1, G-mean, Accuracy/Error, Kappa	5×2 CV	Friedman, Iman-Davenport, Hochberg, t-test	B
SMOTE-IPF (Ref.)	2015	B	9	2	C4.5/C5.0	AUC	rep. 5-CV	Wilcoxon signed-rank, Friedman aligned ranks, Hochberg	A
SplitBal (Ref.)	2015	B	46	2	Naive Bayes, C4.5/C5.0, Rule learner, Random Forest, SVM, k-NN	AUC	rep. 10-CV	Wilcoxon signed-rank, Friedman, Nemenyi	B
A-SUWO (Ref.)	2016	B	16	2	SVM, Logistic Regression, k-NN, LDA	F-measure, G-mean, AUC, Precision, Recall/TPR	rep. k-CV	Friedman, Holm	B
CGMOS (Ref.)	2016	B	30	2	Other, k-NN, SVM, MLP/NN, Random Forest, Boosting ensemble	Precision, Recall/TPR, F1, G-score (P,R), AUC, ROC curve	rep. 10-CV	Wilcoxon signed-rank	A
k-means SMOTE (Ref.)	2018	B	80	2	Logistic Regression, k-NN, Gradient boosting	G-mean, F1, AUPRC	rep. 5-CV	Friedman, Holm	A
OBU (Ref.)	2018	B	36	2	Random Forest	Recall/TPR, Balanced accuracy	hold-out	Wilcoxon signed-rank	B
G-SMOTE (Ref.)	2019	B	63	2	Logistic Regression, k-NN, CART/DT, Gradient boosting	AUC, F1, G-mean	rep. 5-CV	Friedman, Holm	A
RBO (Ref.)	2019	B	34	2	CART/DT, k-NN, SVM, Naive Bayes	Precision, Recall/TPR, F-measure, AUC, G-mean	5×2 CV	Friedman, Shaffer	B
AB-SMOTE (Ref.)	2020	B	8	2	C4.5/C5.0	Recall/TPR, F1	5-CV	none	A
MBS (Ref.)	2020	B	13	2	Logistic Regression, Boosting ensemble	AUC	rep. hold-out	Wilcoxon signed-rank	A
NBU (Ref.)	2020	B	26	2	SVM, Random Forest	Recall/TPR, G-mean, Precision, F1	hold-out	Wilcoxon signed-rank	B
SPE (Ref.)	2020	B	5	2	k-NN, CART/DT, SVM, MLP/NN, Boosting ensemble, Random Forest, Gradient boosting, Logistic Regression, C4.5/C5.0	AUPRC, F1, G-mean, MCC, Recall/TPR, Precision, Runtime	hold-out	none	A
HDE (Ref.)	2021	B	42	2	C4.5/C5.0	AUC, G-mean	rep. 5-CV	Friedman, Holm, Wilcoxon signed-rank	A
FW-SMOTE (Ref.)	2022	B	42	2	k-NN, SVM, Logistic Regression	AUC, G-mean	10-CV	Friedman, Iman-Davenport, Holm	B
GDO (Ref.)	2022	B	52	2	SVM	AUC, G-mean, Runtime	10-CV	Wilcoxon signed-rank	A
KNNOR (Ref.)	2022	B	19	2	SVM, k-NN, CART/DT, MLP/NN	AUC, F1, G-mean	rep. 5-CV	none	A
SVDD-WSMOTE (Ref.)	2022	B	31	2	MLP/NN, LDA, SVM	G-mean, F-measure, AUC, Runtime	rep. 5-CV	Friedman, Holm	A
CISO (Ref.)	2025	B	50	2	Logistic Regression, SVM, Random Forest, Gradient boosting	AUPRC, AUC, G-mean, Fidelity/likelihood, Runtime	k-CV	Friedman, Holm, Wilcoxon signed-rank	C

Study	Year	Set.	DS	Classes	Base learners	Measures	Protocol	Test	Out.
GOIO (Ref.)	2025	B	18	2	Naive Bayes, k-NN, CART/DT, SVM, Random Forest	F1, MCC, Accuracy/Error, Runtime	rep. 5-CV	Friedman, Bonferroni-Dunn, Wilcoxon signed-rank	B
MLOS (Ref.)	2025	B	16	2	SVM	AUC, G-mean, Recall/TPR	rep. 5-CV	Bayesian test	A
QC-SMOTE (Ref.)	2026	B	30	2	Gradient boosting	AUC, Macro-F1, Runtime	rep. k-CV	Wilcoxon signed-rank	A
Global-CS (Ref.)	2006	M	14	3-11	C4.5/C5.0	Cost, MAUC	rep. 10-CV	t-test	A
Global-CS (Ref.)	2010	M	14	3-11	C4.5/C5.0, MLP/NN, Boosting ensemble	Cost, MAUC	rep. 10-CV	t-test	B
Fernandez et al. (Ref.)	2013	M	24	3-11	C4.5/C5.0, SVM, k-NN	AvAcc, F-measure	10-CV	Wilcoxon signed-rank	D
SCUT (Ref.)	2015	M	8	3-10	C4.5/C5.0, SVM, Naive Bayes, k-NN	G-mean, F-measure, AUC	10-CV	Friedman	B
MDO (Ref.)	2016	M	20	3-26	C4.5/C5.0, Rule learner, k-NN	MAUC, G-mean, Precision, Recall/TPR, F-measure	rep. 5-CV	Iman-Davenport, Bonferroni-Dunn	B
SMOM (Ref.)	2017	M	27	3-10	C4.5/C5.0, Boosting ensemble, MLP/NN, Naive Bayes	G-mean, MAUC, Recall/TPR, Precision, F-measure, Per-class recall, Runtime	rep. 10-CV	Wilcoxon signed-rank	B
AMDO (Ref.)	2018	M	17	3-6	C4.5/C5.0	Precision, Macro-precision, MAUC, AUCm, Min-sensitivity	rep. 5-CV	Wilcoxon signed-rank, Friedman, Bonferroni-Dunn	B
Buda et al. (Ref.)	2018	M	3	10-1000	CNN	AUC, ROC curve, Accuracy/Error	fixed split	none	D
GAMO (Ref.)	2019	M	7	5-50 (image); 2-18 (supplement tabular)	CNN	AvAcc, G-mean, Fidelity/likelihood	fixed split	none	A
SOUP (Ref.)	2019	M	15	3-9 (counted from Table 2)	C4.5/C5.0, Rule learner, k-NN	G-mean, Per-class recall, Macro-F1	10-CV	Friedman, Nemenyi, Wilcoxon signed-rank	B
MC-CCR (Ref.)	2020	M	20	3-10	C4.5/C5.0, MLP/NN, k-NN, Naive Bayes	AvAcc, CBA, G-mean, CEN	rep. 10-CV	CV F-test, Friedman, Shaffer	A
MC-RBO (Ref.)	2020	M	20	n/a	C4.5/C5.0, MLP/NN, k-NN, Naive Bayes	AvAcc, CBA, G-mean, CEN	rep. 10-CV	CV F-test, Friedman, Shaffer	B
EMDO (Ref.)	2021	M	11	3-7	C4.5/C5.0	Precision, Macro-precision, AUC, MAUC	rep. 5-CV	none	A
SHSampler (Ref.)	2022	M	20	3-10	CART/DT, Random Forest, Naive Bayes	MAUC, G-mean	5-CV	Friedman, Bayesian test	B
GLOS (Ref.)	2023	M	21	3-11	C4.5/C5.0	G-mean, Precision, Per-class recall, CBA	rep. 5-CV	Wilcoxon signed-rank	B
GMMSampling (Ref.)	2024	M	11	3-9	CART/DT, Naive Bayes, k-NN, Random Forest, Gradient boosting	Balanced accuracy, Macro-F1, G-mean, MCC, AUPRC	rep. 5-CV	Friedman, Wilcoxon signed-rank	B
MC-MBRC (Ref.)	2024	M	20	3-10	Random Forest, CART/DT, MLP/NN, k-NN	AvAcc, F-measure, CBA, CEN	rep. 2-CV	Nemenyi	B
MKC-SMOTE (Ref.)	2024	M	21	n/a	CART/DT, SVM	MAUC, G-mean	5-CV	Wilcoxon signed-rank	A
MC-NCLWO (Ref.)	2025	M	23	n/a	k-NN, CART/DT, Naive Bayes, SVM	F-measure, AvAcc, CBA, CEN	rep. k-CV	Holm	B
SMOTEBoost (Ref.)	2003	B+M	4	2-5	Rule learner	Precision, Recall/TPR, F-measure	10-CV	none	A
COG-OS (Ref.)	2007	B+M	15	2-26	SVM, Logistic Regression, C4.5/C5.0, Rule learner	Precision, Recall/TPR, F1, Accuracy/Error, Per-class recall	hold-out	none	B
DSRBF (Ref.)	2011	B+M	13	2-10	MLP/NN	Accuracy/Error, Min-sensitivity	rep. 10-CV	Friedman, Nemenyi, Bonferroni-Dunn	B

Study	Year	Set.	DS	Classes	Base learners	Measures	Protocol	Test	Out.
CTGAN (Ref.)	2019	B+M	8	binary and multiclass labels (class counts not stated); 1 regression task	CART/DT, SVM, MLP/NN, Boosting ensemble, Logistic Regression	Fidelity/likelihood, F1, Macro-F1, Accuracy/Error	fixed split	none	A
DLexiBoost (Ref.)	2020	B+M	33	2-16	C4.5/C5.0, k-NN	G-mean, AUC, MAUC, Runtime	n.s.	Friedman, Wilcoxon signed-rank, Kruskal-Wallis, Mann-Whitney	B
RSMOTE (Ref.)	2021	B+M	35	2-5	Boosting ensemble, Logistic Regression, CART/DT, k-NN, Gradient boosting, Other	Precision, Recall/TPR, AUC, F1, G-mean, Macro-F1, Macro-precision, Macro-recall	rep. hold-out	Friedman, Iman-Davenport, Holm	B
CBHSID (Ref.)	2022	B+M	16	2-10	SVM	G-mean, Recall/TPR, F1	rep. hold-out	Friedman, Bonferroni-Dunn, Wilcoxon signed-rank	B
SOS (Ref.)	2022	B+M	6	2-6	CART/DT, Logistic Regression, Boosting ensemble, MLP/NN	Weighted-F1, Accuracy/Error, Runtime	hold-out	none	A
NROMM (Ref.)	2023	B+M	30	2-10	C4.5/C5.0, MLP/NN, k-NN, Naive Bayes, SVM	AvAcc, G-mean, CBA, CEN, AUC, Pmin, Macro-precision, AUCm, MAUC	5×2 CV	Friedman, Shaffer, Wilcoxon signed-rank	B
OREM (Ref.)	2023	B+M	49	2-10	C4.5/C5.0, MLP/NN, SVM, CART/DT	F1, G-mean, AUC, Macro-F1, MAUC, Recall/TPR, Precision, Balanced accuracy	rep. 5-CV	Wilcoxon signed-rank	B
GDHS (Ref.)	2025	B+M	20	2-10	CART/DT	G-mean, MAUC	rep. 5-CV	Bayesian test	A
DGOT (Ref.)	2026	B+M	22	n/a	Gradient boosting, k-NN, CART/DT, Logistic Regression, Random Forest	Accuracy/Error, F1, Macro-F1, MCC	rep. 5-CV	Friedman, Bonferroni-Dunn, Wilcoxon signed-rank	A
Loyalty-SMOTE (Ref.)	2026	B+M	59	2-10	SVM	F1, AUC, Recall/TPR, G-mean, Runtime	hold-out	none	A

B = binary, M = multi-class; DS = number of real datasets; k-CV = k-fold cross-validation; rep. = repeated; n.s. = not stated; Out. = outcome class of Table 8.